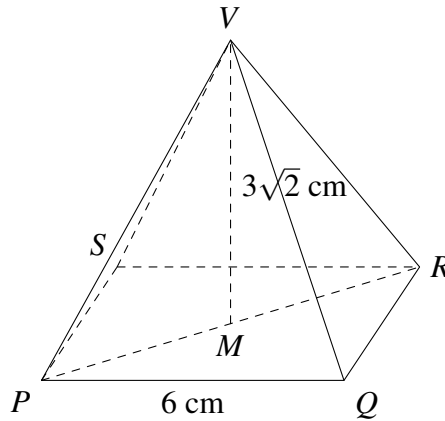


Custom Exam Paper

- 1 The diagram shows a pyramid with a square base $PQRS$.
 The apex V is directly above the centre M of the base.
 The sides of the square base are 6 cm and the vertical height VM is $3\sqrt{2}$ cm.



- (a) Calculate the length PR .

..... [2]

- (b) Calculate the length PM .

..... [1]

- (c) Calculate the angle between the line VP and the plane $PQRS$.

..... [2]

- 2** A cyclist rides a distance of 30 km at an average speed of x km/h. On the return journey, the cyclist increases the average speed by 5 km/h. The return journey takes 30 minutes less than the outward journey.

(a) Show that this information reduces to the equation $x^2 + 5x - 300 = 0$.

[3]

(b) Solve the equation $x^2 + 5x - 300 = 0$ to find the time taken, in hours, for the outward journey.

..... [3]

- 3** (a) Evaluate each of the following.

(i) $81^{\frac{3}{4}}$

..... [1]

(ii) $16^{-\frac{1}{4}}$

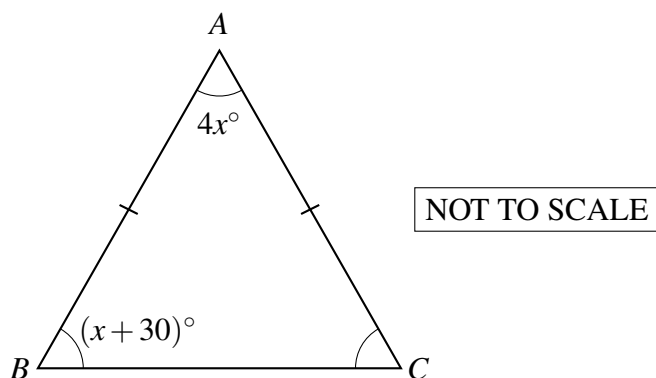
..... [1]

- (b) Simplify the following expression, giving your answer with positive indices.

$$\frac{(2x^3)^4}{4x^5}$$

..... [2]

- 4** In the diagram, ABC is an isosceles triangle with $AB = AC$.



- (a) Find the value of x .

..... [2]

- (b) Find the size of angle BAC .

..... [1]

- 5** Evaluate each of the operations without using a calculator, leaving answers as standard rational values.

(a) $\left(2\frac{7}{9}\right)^{-\frac{1}{2}}$

..... [2]

(b) $\frac{3^{-5} \times 9^2}{27^{-1}}$

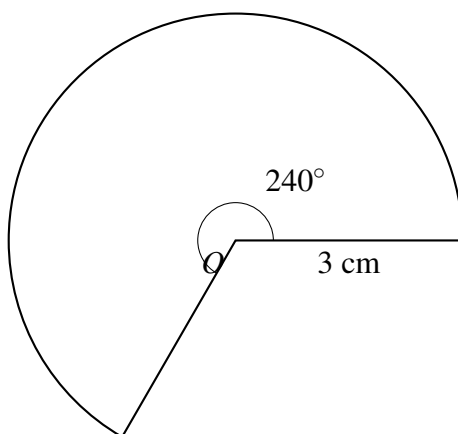
..... [2]

- 6 Make x the subject of the formula.

$$y = \frac{x+5}{x}$$

..... [3]

- 7 A ceremonial badge is shaped as a major sector of a circle with centre O and radius 3 cm. The reflex angle of the major sector is 240° .



- (a) Show that the exact area of the badge is $6\pi \text{ cm}^2$.

[2]

- (b) The badge is coated in liquid silver. The silver coating process costs 5 GBP per $\pi \text{ cm}^2$.

- (i) Calculate the total cost to coat the badge.

..... [2]

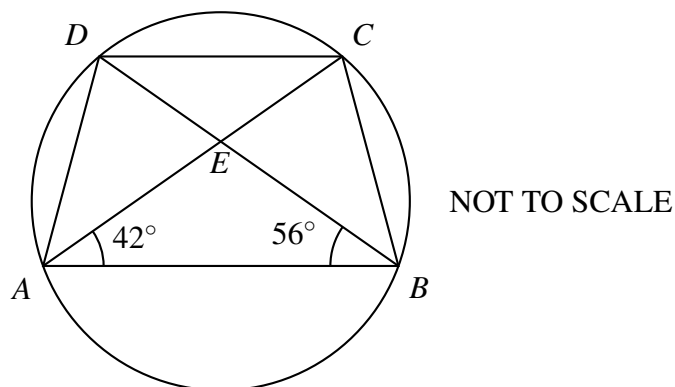
- 8 (a) A cyclist travels the first 12 km of a journey at an average speed of 16 km/h and the remaining 24 km at an average speed of 20 km/h. Calculate the average speed of the cyclist for the entire journey.

..... [3]

- (b) A container has a volume of 4000 cm^3 . It is filled with a liquid that has a uniform density of 1.15 g/cm^3 . Calculate the total mass of the liquid in kilograms.

..... [2]

- 9 Points A , B , C and D lie on a circle.
Chords AC and BD intersect at point E .
Angle $BAC = 42^\circ$.
Angle $ABD = 56^\circ$.



- (a) Find angle ACD .

..... [1]

- (b) Find angle BDC .

..... [1]

(c) Find angle AED .

..... [2]

10 A straight line, L_1 , has the equation $y = -2x + 7$.

(a) Line L_2 is parallel to line L_1 and passes through the point $(3, 4)$. Find the equation of line L_2 in the form $y = mx + c$.

..... [2]

(b) Line L_2 crosses the y -axis at point P . Write down the coordinates of P .

..... [1]

11 The points $A(-3, 8)$ and $B(9, -2)$ define the line segment AB .

(a) Find the coordinates of the midpoint of AB .

..... [2]

(b) Show that the equation of the perpendicular bisector of AB can be written as $6x - 5y = 3$.

[2]

- (c) The perpendicular bisector crosses the x -axis at point Q . Calculate the coordinates of Q .

..... [2]

- 12** Estimate the value of the following calculation by rounding each initial number correct to exactly 1 significant figure. Show your intermediate rounding step clearly.

$$\frac{39.4 \times 4.87}{0.193}$$

..... [3]

- 13** (a) Expand and simplify.

$$3(2x - 5) + 4(x + 2)$$

..... [2]

- (b) Factorise completely.

$$12x^2 + 18xy$$

..... [2]

- (c) Factorise.

$$x^2 - 16y^2$$

..... [2]

- 14** A company purchases a piece of industrial machinery for 50000 EUR. The value of the machinery depreciates exponentially at a rate of 20% each year.

(a) Show that the value of the machinery after 2 years is 32000 EUR.

[2]

(b) Calculate the value of the machinery after 3 years.

..... [2]

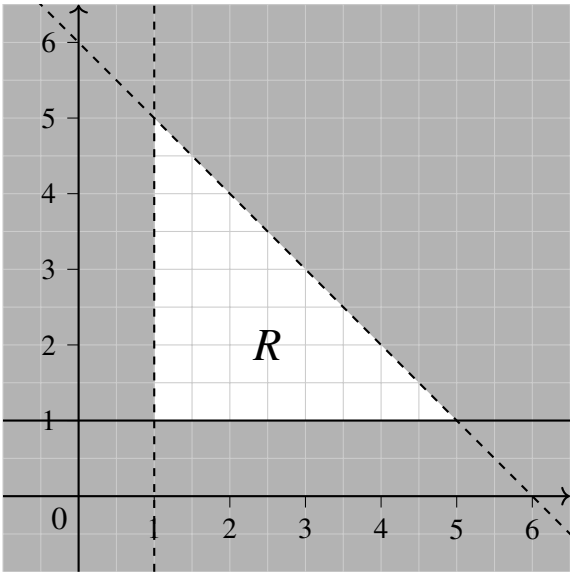
- 15** (a) Convert 12.5% into a fraction written in its simplest form.

..... [1]

(b) Write the mixed number $2\frac{3}{5}$ as a decimal number.

..... [1]

16 The diagram shows a region R , which is unshaded.



Find the 3 inequalities that define the region R .

..... [3]

17 Two similar geometric cones have surface areas of 18 cm^2 and 32 cm^2 .
The height of the smaller cone is 6 cm.
Calculate the height of the larger cone.

..... [3]

18 The line segment joining $C(-2, -5)$ and $D(4, 3)$ forms the diameter of a circle.

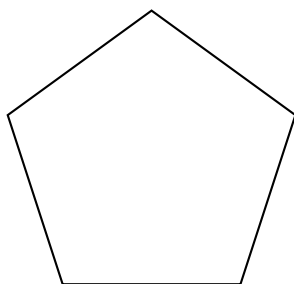
(a) Find the coordinates of the centre of this circle.

..... [2]

(b) Show that the radius of this circle is 5.

[2]

19 Here is a regular pentagon.



(a) Draw all the lines of symmetry on the pentagon.

[2]

(b) State the order of rotational symmetry of this pentagon.

..... [1]

20 (a) Solve the following inequalities algebraically.

(i) $\frac{x+1}{2} < 2$

..... [2]

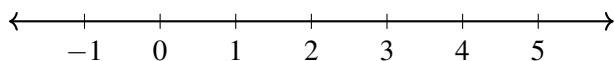
(ii) $-1 \leq 2x + 3 < 9$

..... [2]

(b) List all the integer values that satisfy the inequality in part (a)(ii).

..... [2]

(c) Represent the solution to part (a)(i) on the number line below.



[1]

- 21** A large cylindrical water tank has a cross-sectional area of 2.5 m^2 . Water is poured into the tank at a constant rate of 150 ml per second.

(a) Express the flow rate of the water in litres per minute.

..... [2]

(b) Show that the height of the water in the tank increases by 3.6 mm each hour.

[3]

- 22** (a) A horizontal line passes through the point of intersection of the lines $x = -4$ and $y = 7$. Write down the equation of this horizontal line.

..... [1]

(b) A line L passes through the point $A(6, -5)$ and has a gradient of $-\frac{3}{4}$. Find the coordinates of the points where L crosses:

(i) the x -axis,

..... [2]

(ii) the y -axis.

..... [2]

- 23 (a)** Arrange the following real terms in ascending order of magnitude.

$$\frac{17}{25}, \quad 67.5\%, \quad 0.682, \quad \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$$

..... [2]

- (b)** Complete the expression using the correct symbol from the set $\{>, <, \geq, \leq, =\}$.

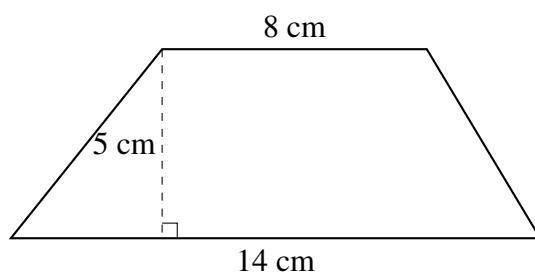
$$3.5\% \quad \text{-----} \quad \frac{7}{200}$$

..... [1]

- 24** A rare collector card is sold for 180 EUR, which represents a profit of 20% on the original purchase price. Calculate the original purchase price.

..... [3]

- 25** The diagram shows a trapezium.



- (a)** Calculate the area of the trapezium.

..... [2]

- (b) A distinct triangle has the same area as the trapezium. The triangle has a base of 11 cm and a perpendicular height of h cm.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Show that the height, h , of the triangle is 10 cm.

[2]

- 26 (a) Evaluate the expression.

$$2 \times \sqrt{81} - 3^2 \times \sqrt[3]{27}$$

..... [2]

- (b) Evaluate the expression.

$$\sqrt{100} \times \sqrt{4} - 5^3$$

..... [2]

27 A specific baking recipe requires 120 g of flour to create a batch of 8 identical scones.

(a) Calculate the mass of flour required to bake exactly 20 of these scones.

..... [2]

(b) Determine the maximum number of full complete scones that can be baked using 450 g of flour.

..... [2]

28 The n th term of an exponential sequence is given by $T_n = 5 \times 2^{n-1}$.

(a) Write down the first three terms of this sequence.

..... [1]

(b) Determine the 6th term, T_6 , of this sequence.

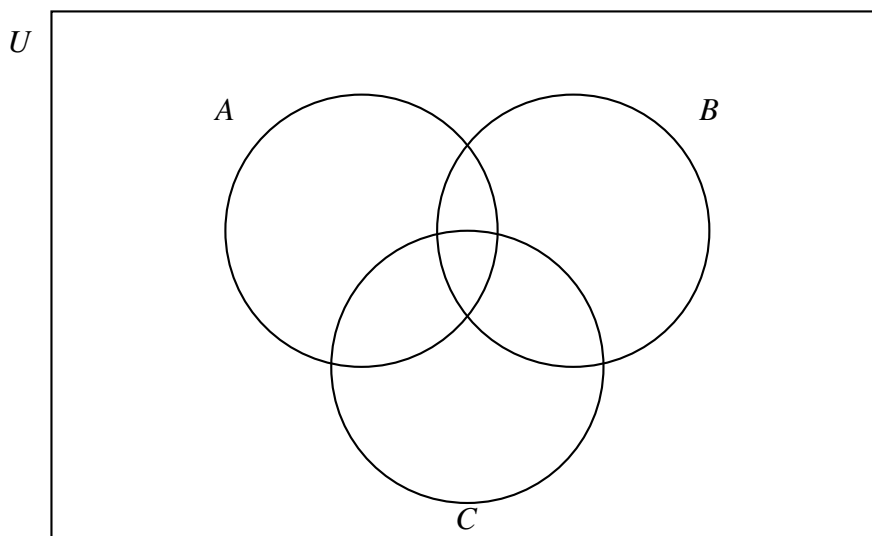
..... [1]

- 29** A group of 100 students are surveyed about their participation in Athletics (A), Badminton (B), and Cricket (C). The results show that 38 students do Athletics, 34 students do Badminton, and 41 students do Cricket. The number of students in the intersection regions are described in terms of x :
- * x students participate in all three sports.
 - * The number of students who do Badminton and Cricket only is $2x$.
 - * The number of students who do Athletics and Cricket only is $x + 1$.
 - * The number of students who do Athletics and Badminton only is $2x - 3$.
 - * The number of students who do Cricket only is equal to the number of students who do none of the three sports.

- (a) Formulate an equation in terms of x for the total number of students and show that it reduces to $155 - 11x = 100$.

[3]

- (b) Complete the Venn diagram to show the exact number of students in each of the 8 regions.



[3]

30 An algebraic fraction equation is given by:

$$\frac{6}{x} - \frac{2}{x+1} = 1$$

(a) Show that this equation simplifies to $x^2 - 3x - 6 = 0$.

[3]

(b) Solve the equation $x^2 - 3x - 6 = 4$.

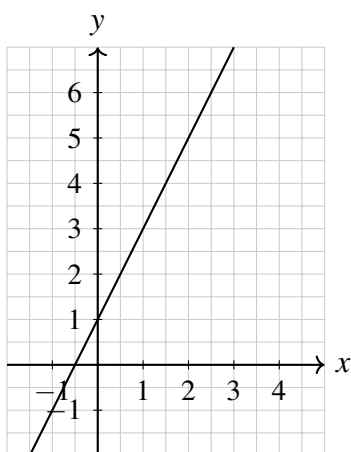
..... [3]

31 Evaluate the expression below, converting the intermediate step out of ordinary units to give your final answer exactly in standard form.

$$(5 \times 10^4) \times (3 \times 10^{-2})$$

..... [2]

- 32** Find the gradient of the straight line shown on the grid below.



..... [2]

- 33** (a) (a) Simplify fully.

$$3\sqrt{48} - \sqrt{12}$$

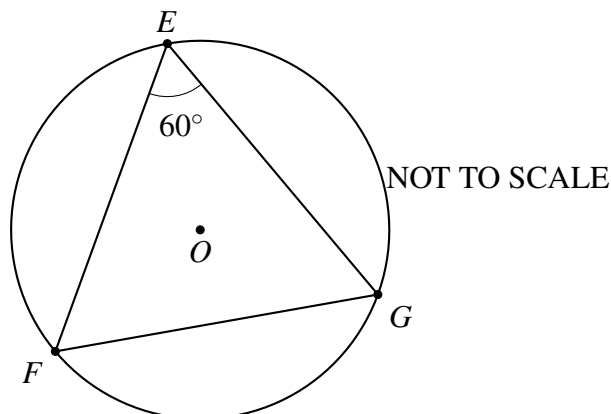
..... [2]

- (b) (b) Rationalise the denominator and simplify fully.

$$\frac{5 + \sqrt{3}}{2 + \sqrt{3}}$$

..... [3]

- 34** The diagram shows a circle, centre O , containing two chords EF and EG . The lengths of the chords are equal, with $EF = EG = 8$ cm. The angle FEG is 60° .



- (a) Write down the exact length of FG .

..... [1]

- (b) Find the exact area of triangle EFG .

$$\text{Area} = \frac{1}{2}ab \sin C$$

..... [2]

- (c) State the geometrical property of circles that ensures the perpendicular distance from O to EF is equal to the perpendicular distance from O to EG .

..... [1]

35 (a) Evaluate.

$$5 \times (3 - 8) + 14$$

..... [2]

(b) Work out.

$$4.5 + 1.2 \div 0.3$$

..... [2]

36 An international cargo ship leaves Port A at 15 20 local time on 10 April. It arrives at Port B after a continuous voyage lasting exactly 3 days and 5 hours. The local time at Port B is 2 hours behind the local time at Port A.

(a) Calculate the local time at Port B when the ship departs from Port A.

..... [1]

(b) Calculate the local time and date at Port B when the ship arrives.

..... [2]

37 A security light flashes every 42 seconds. A second security light flashes every 60 seconds. Both lights flash together at 19:30.

(a) Find the Lowest Common Multiple (LCM) of 42 and 60.

..... [2]

(b) Determine the next time both lights will flash together.

..... [2]